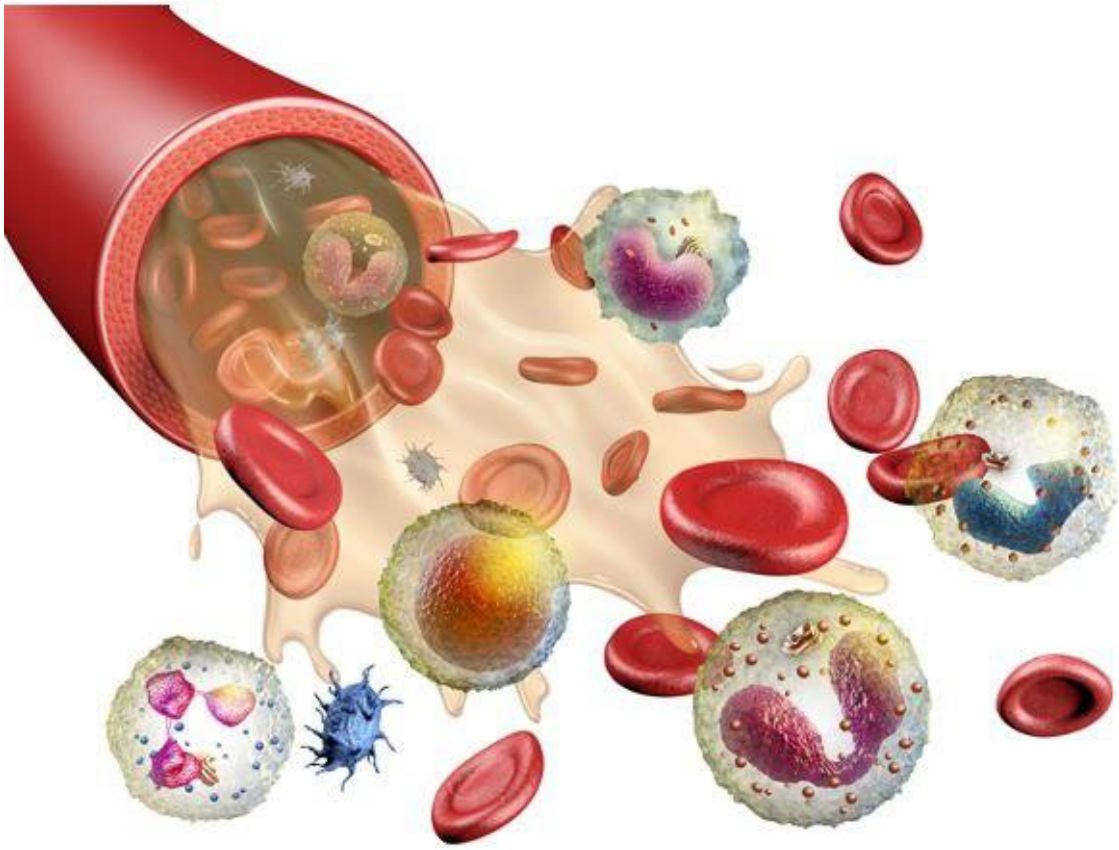
# HAEMATOLOGY



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# Hematopoiesis

- The process which maintains lifelong production of hemopoietin blood cells
- Main site antenatal (yolk sac :1st 8 wks (embryonic Hb )– spleen,liver :8wks–6months (HbF)
- The main site of hematopoiesis throughout postnatal life is the bone marrow.
- Pluripotent stem cells
    - Erythroid cells (RBCs)
    - Myeloid cells (WBCs)
    - Megakaryocytic cells (Plt)

## Hemoglobin:

- RBC contains 640 million molecules of Hb.
- Hb is composed of 4 polypeptide chains
- Hb F (Fetal) : 2 α chains – 2 γ chains (Gamma).
- Hb A (adult) : 2 α chains – 2 β chains.
- Hb $A_2$ : 2 α chains – 2 δ chains (Delta).

## Proportion of Hb Types: (Hb electrophoresis)

Normal Hb at birth:– Hb A (25%) , Hb F (74%),Hb $A_2$ (1%) .mcq

| | HbA | $HbA_2$ | HbF | HbS |
|---|---|---|---|---|
| Newborn | 25% | 1% | 74% | – |
| Adult | 97% | 2% | – | – |
| β-Thalassaemia trait | >90% | ↑ | ↑ | – |
| β-Thalassaemia major | – | ↑ | ↑ | – |
| Sickle cell trait | ✓ | ✓ | ↑ | ✓ |
| Sickle cell disease | – | ✓ | ↑ | ✓ |

## Investigation:

CBC

- Hb – According to age and sex

The Hb concentration is high at birth (>14g/dL) but falls to its lowest level at 2 months of age.

Anemia is defined as a Hb below the normal range. The normal range varies with age, so anemia can be defined as:

• neonate: Hb less than 14 g/dl

• 1 month to 12 months of age: Hb less than 10 g/dl

• 1 year to 12 years of age: Hb less than 11 g/dl

- RBC – 4 to 6 million / $mm^3$
- WBC – 4,000 to 10,000 ( → 10.000 – 20.000/mm3 or more is normal in newborn) : – (if high Leukocytosis & if low Leukopenia)
- Differential: – Absolute count = TLC x percentage of cells
    - Neutrophils
    - Lymphocytes
    - Monocytes
    - Basophils
    - Eosinophils
- Plt– 150,000 to 450,000/$mm^3$ (if high thrombocytosis & if low Thrombocytopenia)
- Ht ( Packed Cell volume (PCV) 35–45%
- MCV 70–100 Femtoliter (mean corpuscular volume)الدم الواحدة متوسط حجم كرة
    - If < 70 = Microcytic RBCs
    - If > 100 = Macrocytic RBCs
- MCH (mean corpuscular Hb)متوسط كمية الهيموجلوبين في كرة الدم الواحدة
    - 27–36 picogram
    - If low = Hypochromic RBCs
    - If high = Hypochromic RBCs
- MCHC: Mean Corpuscular Hemoglobin Concentration تركيز الهيموجلوبين في كره دم واحدة
    - MCHC = 32–36%
    - MCHC = Hypochromic RBCs → Hereditary Spherocytosis

Reticulocyte count:( Mirror of BM activity) Normal < 1%

> 2% Reticulocytosis (Abnormal)

- Hemolytic Anemia
- Response to iron therapy in IDA
- Acute hemorrhage

indices of Hemolysis

- Reticulocytosis.
- Bilirubin (Indirect):– high
- LDH: high
- Haptoglobin: low

**Blood smear (Film)**
**"RBC –Plt – WBC" morphology**

**RBC:**




- **Mentzer's index = MCV / RBC's count** :normal in Thalassaemia (< 13) & high in Iron deficiency anemia (> 13)
- **Red cell Distribution Width (RDW)**:Measure the degree of variation of red blood cell size (Anisocytosis)

# ANEMIA

**Definition: –** It is reduction in hemoglobin concentration or hematocrit below normal for age & sex.

Anemia is an inadequate level of hemoglobin

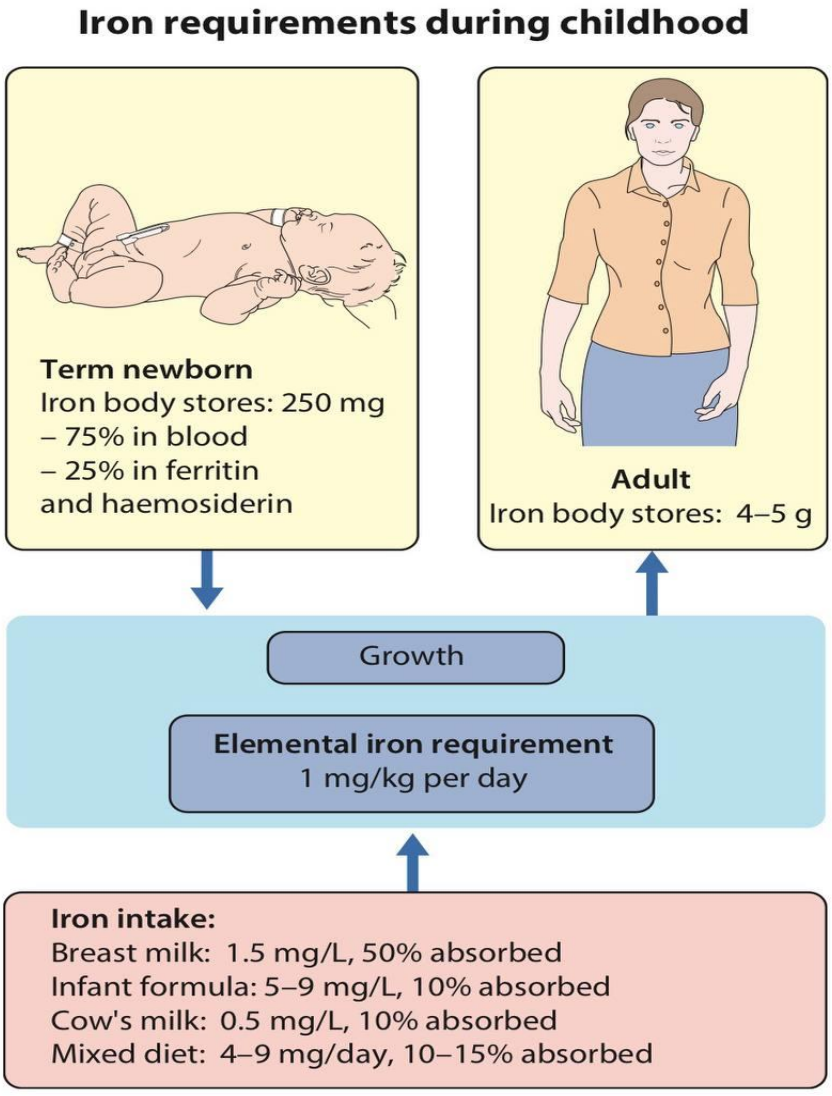
- Neonate: – Hb < 14 g/ dl
- 1–12 M :– Hb < 10 g/dl
- 1–12 Y :– Hb < 11 g/dl

**Causes: –**

| 1. Decrease production | 2. Increase destruction "Hemolysis" |
|---|---|
| • Iron deficiency Anemia<br>• Mega Anemia "$B_{12}$ decrease" and "folate decrease"<br>• Aplastic Anemia "Congenital acquired"<br>• Anemia of chronic disease "CKD"<br>• Bone marrow infiltration (Leukemia)<br><br>Other cause of anemia: Blood loss as Chronic GI Bleeding as (Meckel diverticulum) or Inherited bleeding disorders as (von Willebrand disease) | i. Intracorpuscular (RBC Problem)<br>• **Cell membrane detect**<br>  ○ Hereditary spherocytosis<br>  ○ Hereditary elliptocytosis<br>• **Cell enzyme defect**<br>  • G6pD<br>  • Pyruvate kinase anemia decrease<br>• **Hb defect**<br>  • Thalassemia α – β<br>  • Sickle cell anemia<br><br>ii. Extracorpuscular:–<br>➢ Immune mediated<br>  • Haemolytic disease of newborn<br>  • Autoimmune haemolytic anemia<br>➢ Nonimmune mediated<br>  • Micro angiopathic haemolytic anemia "MAHA"<br>  • Sepsis<br>  • Drugs |

# IRON DEFICIENCY ANEMIA

- Is the **most common** anemia in pediatric
- The most common cause of IDA is **nutritional**
- **Age**: any age "peak 6m – 24 months"
- **Requirements**: 10 mg/day → 10% absorbed → 1 mg/day.
- **Absorption** in: proximal small intestine, mainly Duodenum
- **Distribution**: carried on transferrin (plasma protein, can carry 3 folds than average iron to be fully saturated) → Total Iron Binding Capacity (TIBC)
- **Storage**: Ferritin
- Function: Formation of hemoglobin 70% (Marrow) & myoglobin (Muscles) & Formation of Mitochondria enzymes as catalase.



## Cause of IDA:

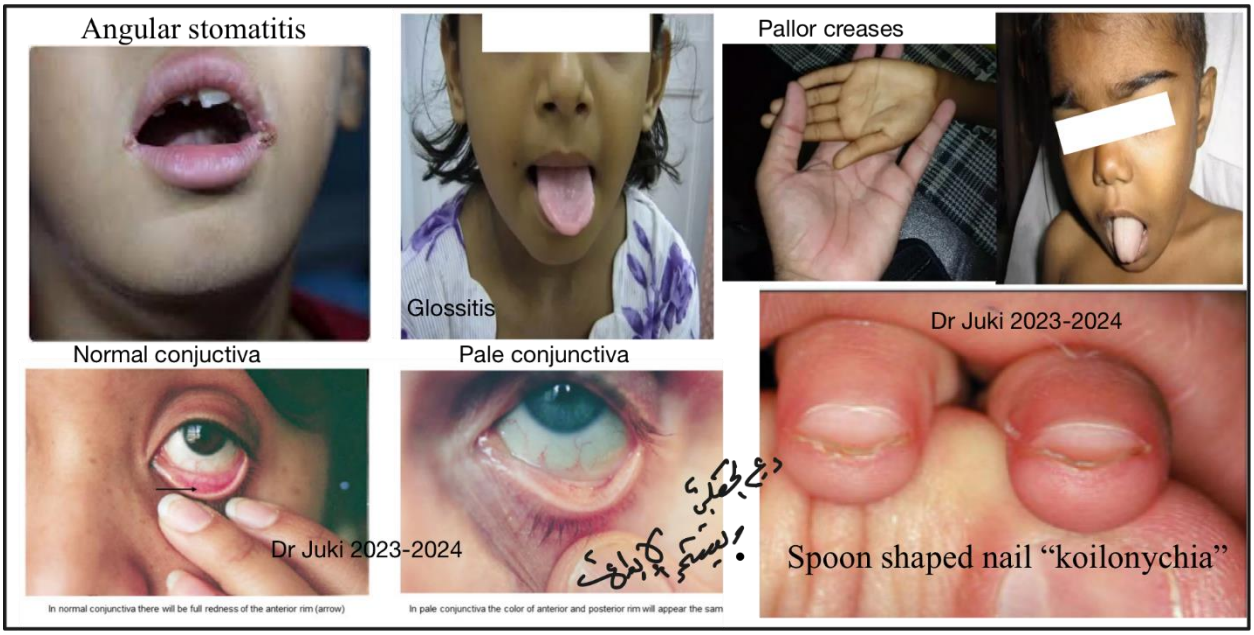
| 1. Decrease intake | 2. Malabsorption | 3. Blood loss |
|---|---|---|
| i. Prolonged exclusive breastfeeding > 6m without supplementation (Breast milk iron low but 50% absorption)<br><br>ii. Cow's milk feeding (higher iron content than breastmilk but only 10% is absorbed))<br><br>iii. Diet deficient in Iron – as unmodified cow's milk.<br><br>**Box 23.1** Dietary sources of iron<br>***High in iron***<br>• Red meat – beef, lamb<br>• Liver, kidney<br>• Oily fish – pilchards, sardines, etc.<br>***Average iron***<br>• Pulses, beans, and peas<br>• Fortified breakfast cereals with added vitamin C<br>• Wholemeal products<br>• Dark green vegetables – broccoli, spinach, etc.<br>• Dried fruit – raisins, sultanas<br>• Nuts and seeds – cashews, peanut butter, etc.<br>***Foods to avoid in excess in toddlers***<br>• Cow's milk<br>• Tea: tannin inhibits iron uptake<br>• High-fibre foods: phytates inhibit iron absorption. | i. Celiac<br><br>ii. CMPI | i. GERD – Gastritis<br><br>ii. CMPI<br><br>iii. Meckel's diverticulum<br><br>iv. Heavy menstrual cycle in adolescent female.<br><br>v. Ankylostoma duodenale (Hookworm Parasite) |

## Clinical picture:

- Usually asymptomatic in most of cases
- Pallor – Decrease feeding – fatigability – breathlessness
- H/O pica: a term which describes the inappropriate eating of non-food materials such as soil, chalk, gravel, or foam rubber.
- Headache – Dizziness – Palpitation – fainting "older child"
- Decrease concentration – behavior – intellectual function"

## Signs:

- Pallor.
- Spoon shaped nail "koilonychia"
- Angular stomatitis – smooth glossitis.
- Tachycardia – flow murmur (ESM).
- Slight splenomegaly (In 10–15% of cases,not common in children clinically )




## Diagnosis:

1. CBC – microcytic hypochromic anemia (Low MCV, MCH, MCHC), reactive Thrombocytosis may present.

2. Blood film – microcytes– Anisocytosis – poikilocytosis – target cells

3. Iron study

- Serum iron: Low
- Serum ferritin: Low
- Total iron binding: high

4. Investigation for underlying cause as

- Stool analysis: – occult BI
- TTG, Endoscopy

**Treatment:** (includes dietary advice & iron therapy)

1. Oral Iron

- Ferrous gluconate syrup or Ferrous sulfate syrup (The best tolerated preparations are Sytron (sodium iron edetate) or Niferex (polysaccharide iron complex) – unlike some other preparations these do not stain the teeth.)
- Dose: 3–6 mg/kg / day
- Duration for 3–6 months (iron therapy should be continued for at least 3 months to replenish the iron stores.)
- S/E (GI upset, Constipation and dark stool)

Response to iron therapy:

| Time after Iron therapy | Response |
|---|---|
| 12–24 hrs | Replacement of intracellular enzymes, improvement of concentration and increase appetite |
| 36–48 hrs | BM response (hyperplasia) |
| 48–72 hrs | Reticulocytosis |
| 4–30 days | increase Hemoglobin (With good compliance, the Hb will rise by about 1 g/ dl per week) |
| 1–3 months | Repletion of stores |

2. **Parenteral iron:** – Indicated in cases of malabsorption, intolerance to oral iron & non–compliance .(**Not more rapid. Not more effective**)

3. **PRBCs Transfusion:** – Only indicated if their very sever IDA "unstable pt"

## Differential diagnosis & Laboratory tests in Microcytic Hypochromic anemia

| MCQ | 1–I.D.A | 2–Anemia of chronic disease | 3–B–Thalassaemia | 4–Sidroblastic |
|---|---|---|---|---|
| 1. Serum Iron | Low | Low | Normal or high | High or N |
| 2. Serum ferritin | Low | Normal or High | Normal or High | High or N |
| 3. TIBC | High | Low | Normal or low | Normal |
| 4. Marrow iron stores | Absent | Present | High in nucleated RBC | Present |

**5–chronic lead poisoning** (Basophilic stippling, high blood Lead )

# MEGALOBLASTIC ANEMIA

Anemia with **megaloblasts** in the bone marrow & **macrocyte** in the peripheral blood due to Folic acid or $B_{12}$ deficiency.

## Folic acid:

- Found in given vegetable – liver – kidney
- Body stores 2–4 months
- Site of absorption :–small intestine

### Causes of Folate deficiency

- Low intake "Goats milk"
- Low absorption "Celiac disease"
- High utilization – Preterm, Hemolysis and Malignant disease
- Antifolate drugs – Phenytoin, Methotrexate and Trimethoprim

## Cyanocobalamin $B_{12}$:

- Found in animal sources only (liver –fish – meat– dairy produce)
- Body stores 2–3 years
- Site of absorption:– terminal ileum (Combine with intrinsic factor (IF) in stomach to Form IF–cobalamin complex)
- Function :Activation of folic acid & Myelination of nervous system

### Causes of $B_{12}$ deficiency:

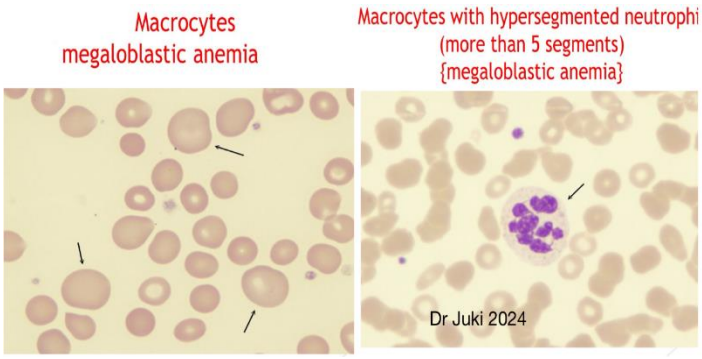
- Low intake "Vegans"
- Low absorption "Crohn's disease" or ileal resection
- pernicious anemia (autoimmune antibodies against parietal cells or IF)
- Chronic liver diseases

## Clinical Picture:

- Common feature of anemia
- Brain manifestations (3P): Late manifestation
    - Peripheral neuropathy
    - Pyramidal tract lesion
    - Posterior column degeneration → sensory ataxia
    - ↓ $B_{12}$ leads to ↓ myelin sheath synthesis → Subacute Combined Degeneration SCD , (**No neurological manifestations in isolated folic acid deficiency**)

## Investigation:

- **CBC** – macrocytic anemia (some cases may they have **pancytopenia** why? )
- B1 film – Macrocytic RBCs & Large hypersegmented neutrophils .
- Serum level of $B_{12}$ :Low .
- Folate (Serum folate level < 4 ng/ml (normal 6–20 ng/ml)
- RBC's folate level (better)




## Treatment:

- Folate oral supplement, 5 mg/day, duration at least 1 month till hematological improvement
- Note :– folate treatment only worsen SCD of vitamin B12 deficiency.
- $B_{12}$ (Hydroxy cobalamin) 1000 pg/monthly, duration for life.
- $R_x$ underlying cause

**Differential Diagnosis of Macrocytic anemia:– MCQ**

1. Megaloblastic anemia (B12 & Folic deficiency)
2. Aplastic anemia as fanconi anemia
3. Chronic liver diseases
4. Bone marrow arrow infiltration (MyeloDysplastic Syndrome MDS)
5. Others : hypothyroidism , down syndrome ,Lesh Nyhan syndrome.

# Hemolytic anemia

## 1–Intracorpuscular Hemolytic Anemia

- Cell enzyme defect
  - G6pD
  - Pyruvate kinase anemia decrease
- Cell membrane detect
  - Hereditary spherocytosis
  - Hereditary elliptocytosis
- Hb defect
  - Thalassemia α – β
  - Sickle cell anemia

## GLUCOSE 6 PHOSPHATE DEHYDROGENASE DEFICIENCY

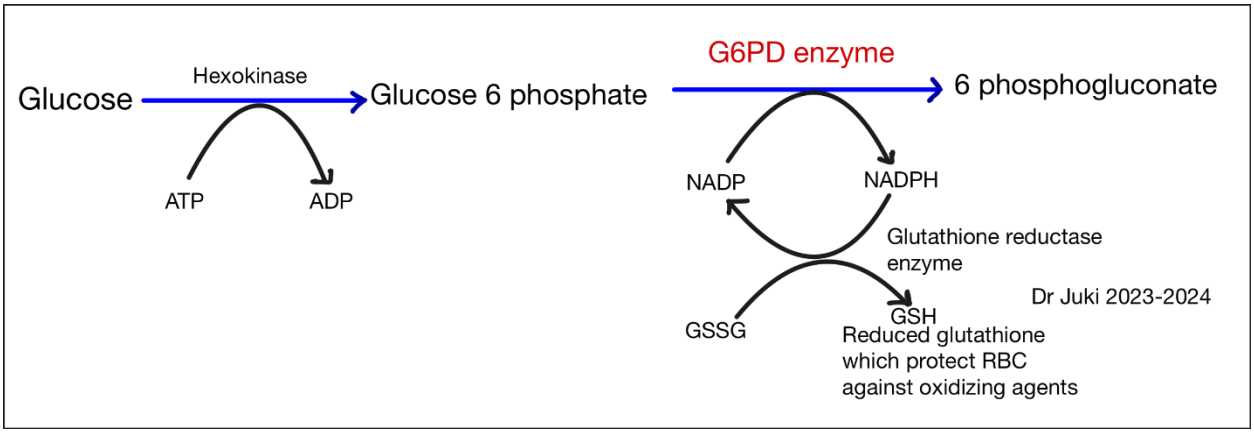
G6pD deficiency is the **most common enzymatic disorder in RBCs**

It's XLR disorder (♂ affected – ♀carrier)

Pathogenesis: deficiency of G6pD which is part of pentose monophosphate pathway

(HMP shunt defect—>↓↓ G-6-PD enzyme—> ↓↓ NADPH production—>

↓↓ Reduction of H2O2 —>Hb oxidation, denaturation & precipitation—> Heinz bodies —–> increase RBC destruction.)



**Oxidizing agents "Triggers"**

- Food "**Fava** beans" Favism أنيميا الفول
- Infection viral or bacterial. " " **viral infection is most common triggers** .
- Chemicals
    - Naphthalene (mothballs)
    - Divicine (fava beans – also called broad beans)
- Drugs as
    - Antibiotics: Nitrofurantoin – sulfonamides – Dapsone ciprofloxacin – Quinolones
    - Analgesia: Aspirin (high dose)
    - Antimalarials : Primaquine chloroquine
    - Others: Vit k –phenylhydrazine

## Clinical Picture:

1. The majority of cases may remain clinically **asymptomatic**.

2. **Neonatal jaundice** (1st 24 hours or days – severe needs exchange transfusion) may lead to kernicterus (Indirect hyperbilirubinemia) .

3. **Acute hemolytic** anemia

- Episode of hemolysis occurs in exposure to oxidant stress as medication – chemicals – after ingestion of fava beans
- Usually begins 24–72 **hours** after exposure to oxidant stress
- C/P: malaise – irritability – pallor – jaundice –tachycardia , dark urine– abdominal pain – high or low fever.

## Investigation:

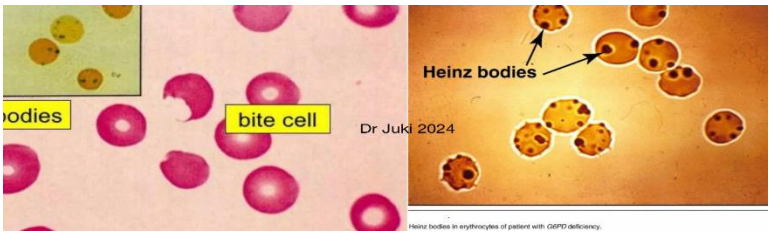
1. CBC:– normolytic normochromic anemia

2. Indices of hemolysis

- Reticulocyte count > 2% (high)
- LDH: high
- Indirect bilirubin: high
- Haptoglobin: low

- Hemoglobinuria (intravascular hemolysis)

3. Peripheral blood smear (Heinz bodies – Bite Cells – blister cells)




4. **Enzyme level assay (semi-quantitative – Quantitative tests)**

Mcq (should be done or repeated 2-3 months after acute attack to avoid false -ve result due to higher enzyme concentration in new RBC "reticulocytes")

5. **Genetic test** (for family screening – prenatal diagnosis)

6.**Neonatal screening is available in some countries MCQ .**

**Treatment:** Acute hemolysis is usually **self-limiting**, often resolving after **8 to 14 days,** transfusion is needed in cases of **severe** anemia.

1. Stop oxidant agents

2. PRBCs transfusion (if severe anemia)

**Prevention:**

1. Acute hemolysis in G6pD patients is largely preventable (so parents should be given advice about the signs of acute hemolysis (jaundice, pallor, and dark urine) and provided with a list of drugs, chemicals, and food to avoid) .

2.Persons with chronic or recurrent severe hemolysis should be placed on daily folic acid supplements. Consultations with a hematologist are ideal for long-term follow up

**Pyruvate kinase deficiency:**

- AR – most common enzyme deficiency in glycolysis pathway in RBCs
- ↓↓Pyruvate kinase enzyme—>↓↓ ATP production —> ↓↓RBC life span.
- C/P – Neonatal jaundice and late childhood with chronic hemolysis – splenomegaly
- Rx – PRBCs transfusion and splenectomy.

# Hereditary Spherocytosis "HS"

- **Autosomal dominant** AD [But 25% No F/H & new mutation]
- % 1: 5000 Births in Caucasians

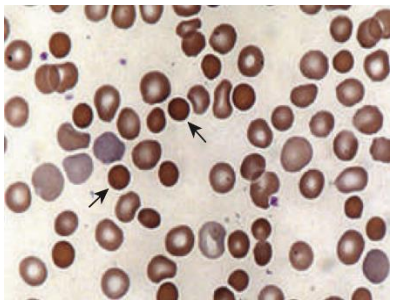
## Pathogenesis:

In HS : intrinsic defect in erythrocyte membrane proteins (Spectrin – Ankyrin– band 3) result in RBC cytoskeleton instability —>RBC become spherical in shape– less flexible –less deformable & rapidly removed from circulation spleen "Hemolysis"

## Clinical Picture: "F/H of HS may be present"

1. Neonatal Jaundice "indirect", severe .
2. Chronic hemolysis * S/S Anemia – Jaundice "Intermittent"
   *Splenomegaly:– mold to moderate "6 m – 1 yr"
   *Gallbladder stone: –due to increase bilirubin excretion
3. A plastic crisis:– Uncommon, transient, caused by parvovirus $B_{19}$

## Investigations:



1. CBC: Normocytic Anemia , high MCHC.
2. Reticulocyte count High >2%, high indirect bilirubin & LDH.
3. Coomb's test "–ve"
4. Blood Film: show spherocyte RBC.
5. Osmotic fragility test "Diagnostic" +ve (RBCs are placed in hypotonic saline): in HS: hemolysis starts at higher concentrations unlike normal RBC. (i.e., more fragile)
6. Dye binding assay by flow cytometry " EMA test " (new test ).
7. Genetic test

## Treatment:

1. Most children with mild chronic hemolytic anemia require only (Folic Acid + Follow Up).
2. In severe Anemia: PRBCs transfusion
3. **Splenectomy** (open or Laparoscopic)
   - **Indicated if severe hemolysis or troublesome symptoms of anemia eg "severe tiredness – loss of vigor" or poor growth**
   - **Done after age of 6 years "B/C risk of sepsis"**
   - **Before splenectomy vaccination [Hib – meningococcal–pneumococcal vaccine are indicated**
   - **After Splenectomy Penicillin Orally indicated lifelong.**
4. Aplastic crisis from parvovirus B19 infection usually requires one or two blood transfusions over the 3–to–4–week period when no red blood cells are produced.
5. If gallstones are symptomatic, cholecystectomy may be necessary

# Thalassemia

## α Thalassemia

- AR
- Defect in α chain synthesis
- α chain gene present in chromosome *16*
- 4 α globin genes
- The manifestation of α–thalassemia syndromes depends on the number of functional α–globin genes.

**Types:**

1. **Silent carrier or trait** [α– / αα] "Normal CBC"
2. **α thalassemia trait** [αα /––] "Mild microcytic hypochromic anemia" may be confused with IDA.
3. **Hemoglobin H** [α–/––] "microcytic anemia + mild hemolysis, blood transfusions.
4. **Hb Barts hydrops fetalis** [––/––] "severe anemia, no HbA (α2β2) – anasarca – HF" Death in utero or early hours of life.

## β Thalassemia

- AR
- Defect in beta chain synthesis
- Beta chain gene present in chromosome *11*
- 2 Beta globin genes

Types:

1. **β thalassemia trait "minor"** "Asymptomatic– mild anemia" No Rx require
2. **β thalassemia intermedia** "mod hemolysis – severe anemia" ,the β–globin mutations allow a small amount of HbA and/or a large amount of HbF to be produced.
3. **β thalassemia major** "severe hemolysis – transfusion dependency – iron overload (most severe form of the disease, HbA (α2β2) cannot be produced )

## β Thalassemia trait (Heterozygotes)

- Usually **asymptomatic** (Anaemia is mild or absent, with a disproportionate reduction in MCH and MCV ).
- **Microcytic Hypochromic anemia**(misdiagnosed as IDA) ,but Serum Ferritin is "High" Unlike IDA .
- **Hb electrophoresis**(Raised $HbA_2$ most important diagnostic usually about≃ 5%, – mild elevation in Hb F 1–3%)
- $R_x$ – **No treatment is required.**

**To differentiate IDA from B thalassemia by the following:**

1. High RDW>15 In IDA .
2. Mentzer Index =MCV: RBCcount
   i. – If more than 13 =IDA
   ii. – If less than 13 = B thalassemia
3. HB electrophoresis: abnormal in thalassemia.

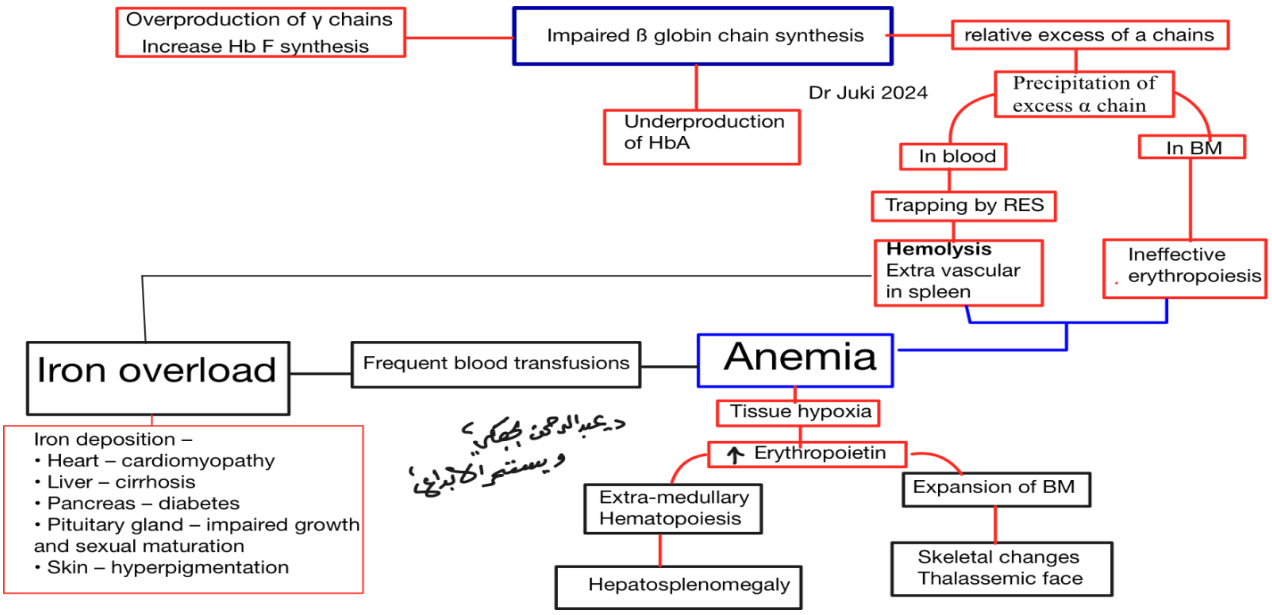
# "β Thalassemia major"

## "Cooley's Anemia"

- The **most severe** form of thalassemia.(**Quantitative**)
- **Homogenous** thalassemia, (2 genes mutation or deletion).
- The commonest chronic hemolytic anemia in the **Mediterranean** and middle east.
- The **commonest chronic** hemolytic anemia **dependent** on blood transfusion.

### Pathogenesis:

1. "Impaired **synthesis**" **of beta chains**
   - decrease production of Hb A —>deposition of excess free α chain inside RBCs —>Hemolysis chronic [Anemia & Jaundice] —> ++ BM hematopoiesis [ BM expansion]– thalassemia face & Extramedullary hematopoiesis (splenomegaly + hepatomegaly).
   - Compensatory production of other chains(gamma —>Increase HbF, delta —>$HbA_2$ increase )
2. **Iron overload** :
   - **Liver** :– Liver cirrhosis
   - **Pancreas** :– DM (Bronze)
   - **Heart**:– Cardiomyopathy
   - **Skin** :– Hyperpigmentation
   - **Pituitary Gland** :– Delayed puberty – FTT

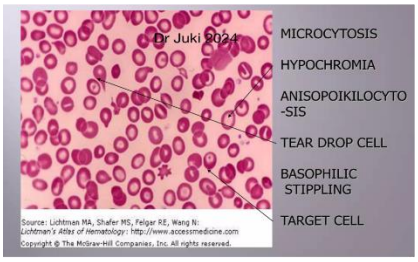



## Clinical Picture:

1. Severe Anemia (onset :– 3–6 **month** of age)
2. Features of **chronic** hemolysis [pallor – Jaundice – Splenomegaly– gallbladder stone]
3. Failure to thrive/growth failure
4. **Thalassemic face**:– bone marrow expansion/ the latter leads to the classical facies with maxillary overgrowth and skull bossing.
5. C/P of **iron overload (Hemosiderosis)**
6. **Crisis: aplastic crisis** by parvovirus $B_{19}$", hemolytic crisis.

## Investigation:

1. CBC: microcytic **hypochromic anemia – reticulocyte count** >2%
2. Indices of hemolysis: reticulocytosis , Increase indirect bilirubin ,increase stercobilinogen in stool ,LDH ,Low haptoglobin.
3. **Increase iron** ,low TIBC.
4. RDW & Mentzer index both are **normal** in thalassemia.
5. Blood Film :**Target** cells – anisocytosis – poikilocytosis –Tear drops
6. **Hb electrophoresis** :Diagnostic (HbA : low and HbF >90% and $HbA_2$ increase )
7. Skull XRay :hair on end no durance



Blood film



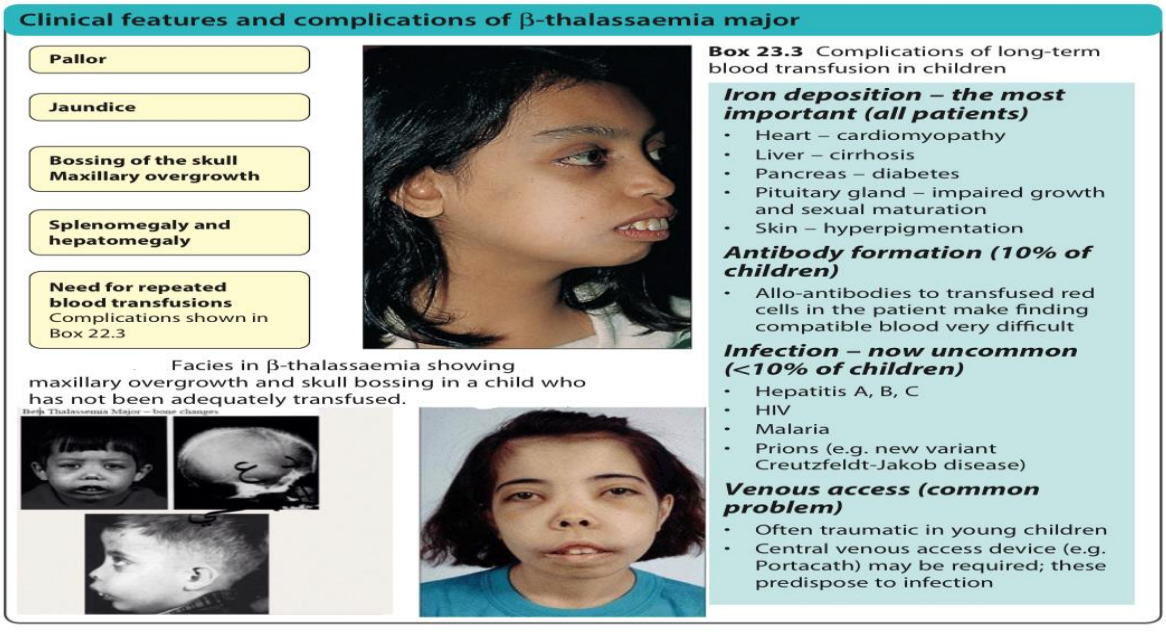
Hair on end appearance

## Treatment:

1. **Blood transfusion**
   - Given monthly to maintain Hb > 9.5–10 g/dL why?
   - Allow normal growth – decrease medullary + extramedullary hematopoiesis
2. **Iron chelation therapy (to treat iron overload)**
   Indication: Serum ferritin>1000 ng/dl ,starting from 2–3yrs age.
   Types: Deferoxamine "SC injection " and Deferasirox "oral "
3. Folic Acid
4. **Hydroxyurea**:– it increased HbF Production
5. **Splenectomy**:if [massive splenomegaly – hypersplenism– increase frequency of transfusion]
6. **Bone marrows transplantation – is curative**
7. **Genetic** counseling is important , prenatal diagnosis of β–thalassaemia is possible by (DNA analysis of a chorionic villus sample).

## Complication & Cause of death:

1. Heart Failure
2. Hypersplenism (splenomegaly +pancytopenia)
3. Iron overload ***
4. Infection transmission (<10%) hepatitis B–C, HIV, Malaria, Prions
5. Antibody formation(10%) allo – Antibodies to transfused red cells
6. Venous access (common problem) "portacath may be required



Facies in β-thalassaemia showing maxillary overgrowth and skull bossing in a child who has not been adequately transfused.

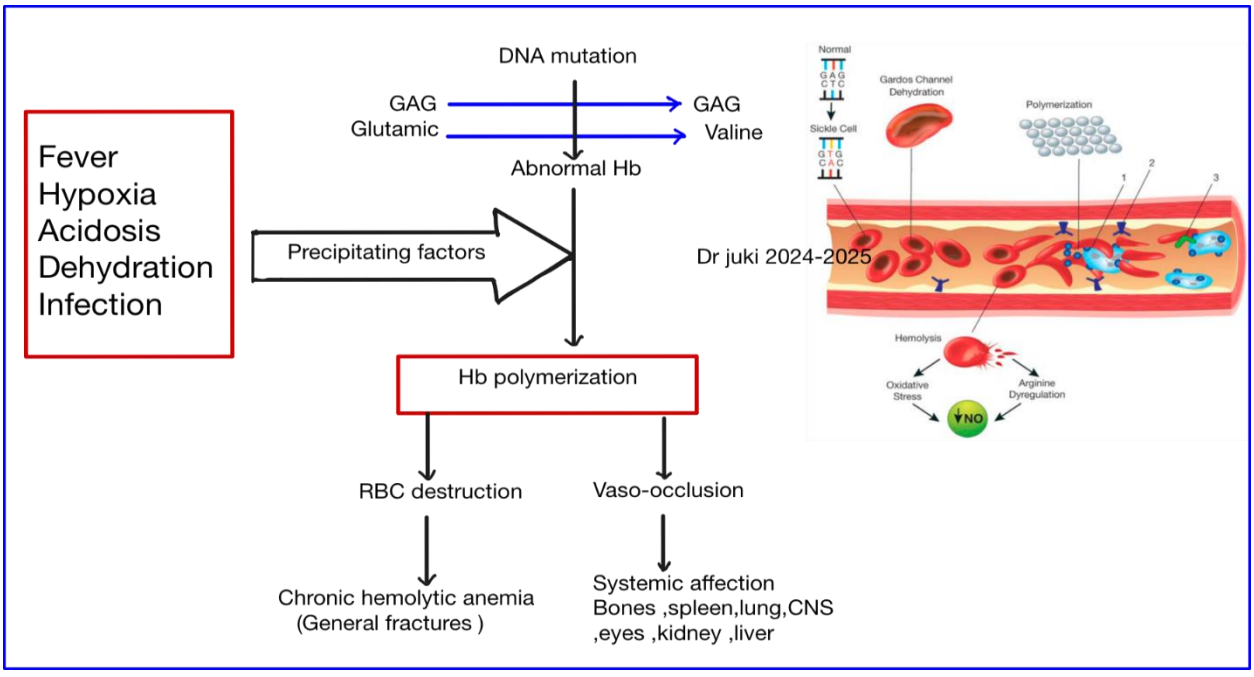
# "Sickle Cell disease"

- **Autosomal Recessive "AR"**
- More common in **black**
- **Pathogenesis:**
- It's structural defect in beta chain (valine replaces glutamic acid the 6th aa from amino end)

## Types:

1. **Sickle cell anemia (HbSS)**, Homogenous form (Abnormal 2 genes) [all their Hb is HbS/ they have small amounts of HbF and no HbA because they have the sickle mutation in both β-globin genes].
2. **Sickle Carriers (Sickle trait)** (HbAS) (Asymptomatic), Heterogeneous form (Abnormal 1 gene) , where there is inheritance of HbS from one parent and a normal β-globin gene from the other parent, ≃ 40% of their hemoglobin is HbS.

## Pathogenesis of SCA

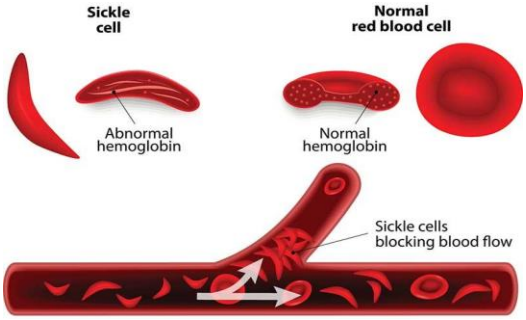



- Hb S with exposure to hypoxia —>Hb Polymerization —>Sickling RBC become sickle shaped —>increase viscosity—>vascular occlusion "Ischemia" and being rigid —>Premature destruction and hemolysis

# Sickle cell anemia: HbSS

## Clinical picture (Usually >6 months of age)

1. General features of **chronic hemolytic anemia**, pallor and Jaundice ,gallbladder stone.
2. Clinical pictures of **crisis**




I. **Vaso – Occlusive crisis (Painful crisis)**

- **The most common crisis and specific for SCA**
- causing pain affects many organs of the body with varying frequency and severity.
- **Predisposing Factors: Hypoxia – Fever – Acidosis – infection –Dehydration –excessive exercise or stress–cold.**
- Manifestations: Due to vascular occlusion and infarction in different sites (The bones of the limbs and spine are the most common sites)

1. **Hand Foot Syndrome (Dactylitis)**
   - Very common <2 years (usually <5 years)– painful swelling of hands and feet



DD: osteomyelitis & compartment syndrome.

2. **Acute chest Syndrome (most serious)**
   - C/P {Sudden breathless and Cough – Fever – RD – high WBC – X Ray shows patches}
   - Can lead to severe hypoxia and the need for mechanical ventilation and emergency transfusion.
3. CNS :– Convulsions, motor deficits(hemiplegia), stroke & blindness.
4. Abdomen :– severe Acute Abdominal pain
5. Kidneys :– Painless Hematuria (due to papillary necrosis)
6. Priapism:– Needs to be treated promptly with exchange transfusion as it may lead to erectile impotence (DD leukemia ).
7. Avascular necrosis of femoral head – leg ulcer.
8. **Spleen**: Frequent crisis leading to **Autosplenectomy "Hyposplenism" increase risk of infection "Capsulated Organism"**

Splenomegaly :– may present in young children, but becomes much less frequent in older children due to autosplenectomy .

II. **Sequestration crisis:**
- Occlusion of venous outflow —>Acute pooling of large amount of blood in the spleen ± liver —>Anemia and massive – Hypovolemic shock– splenomegaly .

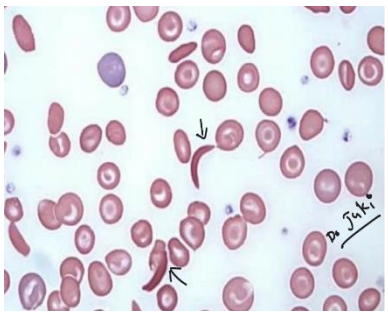
III. **Aplastic crisis:**
- Sudden pancytopenia (Bore marrow hypoplasia)
- Caused usually by parvovirus.

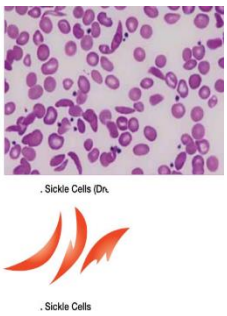
IV. **Hemolytic crisis** :–Hyper hemolysis severe anemia – Jaundice

V. **Megaloblastic crisis** :– Due to folate deficiency .

## Investigation:

1. CBC : Normocytic normochromic anemia (mild to moderate anemia)
2. Retic Count >2% – Indirect bilirubin high
3. Blood film :**Sickle shaped cells**– Howell jolly bodies – Target cells









4. Hb electrophoresis (Diagnostic) – **HbS 80–95% and HbF 2–20% and No HbA** .
5. Neonatal screening test (Guthrie test) & antenatal diagnosis is available .

## Treatment:

- Folic Acid daily .
- Vaso–occlusive crises should be minimized by avoiding exposure to cold, dehydration, excessive exercise, undue stress, or hypoxia.
- Hydroxyurea (hydroxycarbamide) :it increase HbF (increase HB F , not prone to sickling., decrease crisis)

## Treatment of crisis:

- **Treatment Vaso – occlusive crises**
  1. Analgesia for pain [oral or intravenous as required] ,paracetamol, brufen or opiate .
  2. $O_2$ if RD or hypoxia & low O2 saturation.
  3. Correction of dehydration oral or IV (IVF :– 1 ½ M)
  4. Antibiotic empirically for infection .
  5. Exchange transfusions in severe cases (acute chest syndrome, stroke and priapism.
  6. packed cell transfusion, given only in sever symptomatic anemia.

- **Aplastic – sequestration – hyper hemolytic crisis** :– $R_x$ by blood transfusion

Control infections (due to functional asplenia)

- Prophylactic oral **penicillin** throughout childhood (or lifelong )
- Vaccination :– Hib – pneumococcal & meningococcal vaccine

- Bone marrow transplantation (Curable)

## Complications of SC disease :–

1. **Short stature** and delayed puberty
2. **Infection**: streptococcal pneumonia **, osteomyelitis.
3. **Stroke,moyamoya disease** and cognitive problems may manifestes as subtle neurological damage ,often manifest with poor concentration and school performance
4. **Adenotonsillar hypertrophy** –>sleep apnoea syndrome —>nocturnal hypoxaemia, —> vaso–occlusive crises and/or stroke
5. **Cardiac enlargement & Heart failure** – from chronic anemia
6. Renal dysfunction ,CKD– enuresis .
7. **Gallstones** – due to increased bile pigment production.
8. **Leg** ulcers – uncommon in children.
9. **Psychosocial problems** – difficulties with education and behavior .

- **Sickle cell trait (Hb AS)**
  - Asymptomatic
  - Sickling occurs only in severe hypoxaemia
  - Hb S 20–40% and Hb A 60–80%
  - Protection against malaria
- **Sickle Cell – Beta thalassemia (Hb S/ B thalassemia)**
  - Sickle cell gene from one parent and Beta thalassemia gene from other parent
  - Moderate anemia
  - Splenomegaly >5 years
- **HbSC disease(HbSC)–other type : rare**
  - Normal hemoglobin level and fewer painful crises than SC disease
  - Develop proliferative retinopathy in adolescence & osteonecrosis of the hips and shoulders.

# 2- Extracorpuscular Hemolytic Anemia

**Definition:-** ↓↓ RBC life span due to premature destruction caused by extra-corpuscular factors

1. **Immune hemolytic anemia**

a. Active: **Autoimmune hemolytic anemia** (auto-Ab are formed by the patient)

b. Passively acquired antibodies: Isoimmune hemolytic anemia (**Hemolytic disease of the newborn** شرح في شابتر نيونيت (**Rh,ABO,minor**) & mismatched blood transfusion)

2. **Nonimmune hemolytic anemia:** MAHA, infections, hypersplenism, Wilson, drugs

## Autoimmune hemolytic anemia (AIHA) :-

**Definition:-** ↓↓ RBC life span due to premature destruction by auto-Ab formed by the patient

**Classification & Clinical Picture**

| AIHA due **warm** antibodies (common) | AIHA due **cold** antibodies |
|---|---|
| **Etiology**<br>1. **Primary (Idiopathic)***most common<br>2. **Secondary:**<br>• Collagen-vascular diseases(SLE,UC)<br>• Malignancy (lymphoma)<br>• Infections such as hepatitis,CMV,measles,covid19<br>3. Drugs as penicillin, quinine | **Etiology**<br>1-**Primary (Idiopathic)**<br>2. **Secondary**<br>• Malignancy (lymphoproliferative)<br>• Infections (mycoplasma, EBV*)<br>3. **Paroxysmal cold hemoglobinuria**<br>Due to the presence of cold IgG (not IgM). |
| **Autoantibodies**<br>• IgG (monomer)<br>• Active between 35-40° C<br>• Do not require complement for activity (extravascular hemolysis: spleen)<br>• Do not cause agglutination in vitro | **Autoantibodies**<br>•IgM (pentamer)<br>• Active < 37 C (best at 0-4 C)<br>• Require complement for activity(intravascular hemolysis)<br>• Cause agglutination in vitro |
| **Clinical picture**<br>**A). Acute AIHA:**Age= 2-12 years<br>Acute onset of pallor, jaundice & Hb-uria ,Huge splenomegaly | **Clinical picture**<br>Hemolysis & Hb-uria follow cold exposure & mild splenomegaly . |

| | |
|---|---|
| **Good response to steroid**<br>Full recovery within 3–6 months<br>B) **Chronic AIHA**:Age >12 years<br>Gradual onset & splenomegaly | |
| **Investigation**<br>1. CBC :Normocytic ,normochromic anemia<br>2. Indices of hemolysis<br>a. Reticulocyte count > 2% (high)<br>b. LDH : high<br>c. Indirect bilirubin :high<br>d. Haptoglobin : low<br>3. Blood film :spherocyte (imp DD for Hereditary spherocytosis)<br>4. **Coombs test (Direct antiglobulin test:):+ve** (for detection of auto–Ab): Most important | **Investigation**<br>1. CBC :Normocytic ,normochromic anemia<br>2. Indices of hemolysis<br>a. Reticulocyte count > 2% (high)<br>b. LDH : high<br>c. Indirect bilirubin :high<br>d. Haptoglobin : low<br>3. Blood film :spherocyte (imp DD for Hereditary spherocytosis)<br>4. **Coombs test:–+ ve** (for detection of auto–Ab): Most important<br>5. Cold agglutination test (+++) |
| **Treatment**<br>1–Rx of the cause, avoid drugs.<br>2. Blood transfusion (should be avoided ??)<br>3. **Steroids(DOC)**: IV methylprednisolone or prednisone oral.<br>4. IVIG: block splenic Fc receptors.<br>5. Rituximab (anti–CD20):in refractory cases.<br>6.Plasmapheresis: removal of auto–Ab.<br>7. Immunosuppressive.<br>8.Splenectomy (rarely done in children unlike adult & resistant cases ) | **Treatment**<br>1–Rx of the cause, avoid cold.<br>2. Blood transfusion(should be avoided ??)<br>3.IVIG: block splenic Fc receptors<br>4. Plasmapheresis: removal of auto–Ab(better effect in cold).<br>5. Immunosuppressive<br>6. **Rituximab** (anti–CD20):<br>7–Poor response to steroid |

**Note :– Evan syndrome = AIHA + immune Thrombocytopenia**

# "Aplastic Anemia"

It's bone marrow failure(**reduction or absence** of all three main lineages in the bone marrow leading to peripheral blood **pancytopenia** associated with **hypoplastic** bone marrow).

## Causes:

| 1–Congenital inherited | 2–Acquired | 3–Idiopathic** |
|---|---|---|
| • **Fanconi anemia**<br>• Shwachman diamond syndrome | • **Infection (Viral especially hepatitis – Bacteria)**<br>• Chemical as benzene<br>• DrugsasChloramphenicol–sulfonamide –phenytoin – chemotherapy .<br>• Irradiation<br>• Disease as paroxysmal nocturnal hemoglobinuria | • Specific cause cannot be identified. |

## Clinical picture:

1. Anemia due to decreased RBC.
2. Infection due to decrease WBC (neutrophils)
3. Bruising and bleeding due to thrombocytopenia .
4. Usually No lymphadenopathy – No hepatosplenomegaly

## Diagnosis:

1. **CBC** – low Hb – low WBC – low plt – (**reticulocytopenia**)
2. **Blood Film** :Decreased Plt – Decreased RBC – Decreased WBC .
3. **Bone Marrow aspiration** :hypoplastic with Increase fat with No evidence of malignancy

## Treatment:

1. Supportive :– PRBCs – plt transfusion & control infection
2. Stem cell or BM transplantation .

# "Fanconi Anemia"

- Is the **most common inherited** form of aplastic anemia
- Is **Autosomal recessive** " AR" caused by mutations in one of the many FANC genes, most commonly FANCA.
- **Age at onset:** of bone marrow failure usually becomes apparent by the age of 5 years or 6 years.

## Clinical Picture:

**Pancytopenia + Dysmorphic Features & Congenital Anomalies "75%"**

( S الحفظ كلهم بيبدو بحرف لسهولة )

- Short Stature H+<3rd centile
- Small head "Microcephaly OFC <3rd centile "
- Skin Pigmentation "Cafe aliat sports"
- Skeletal anomalies absent or abnormal thumb – Radii
- Small Eyes (microphthalmia)
- Shoe "Horseshoe" or kidney agenesis .
- Septal ventricular defect VSD " CHD "
- Solid tumor & Leukemia % is high .
- Learning disability & Low IQ




## Diagnosis

1. CBC :– Pancytopenia – **high MCV**.
2. **Bone Marrow aspiration** :– Hypoplastic & hypocellular .
3. **Chromosome breakage test** of peripheral blood lymphocytes( used to identify affected family members or for prenatal diagnosis).
4. HbF:– high
5. Genetic study

## Treatment:

1. Supportive cane :–PRBCs – plt transfusion and Control infection (G– CSF)
2. Androgen therapy (50–75% response) + erythropoiesis .
3. Steroid therapy can be used also (S/E) .
4. BM transplantation .

| Schwachman diamond Syndrome | Diamond –Blackfan Syndrome (DBA) |
|---|---|
| • AR<br>• Clinically<br>1. Bone marrow failure (isolated neutropenia or pancytopenia)<br>2. Pancreatic exocrine failure(chronic diarrhea + FTT )<br>• Increased risk of leukemia | • Is called Pure red cell aplasia<br>• AR (F/H in 20%, but 80% are sporadic)<br>C/P:– Anemia in infancy (Most cases present at 2 –3 months of age, but 25% present at birth)<br>• WBC– Plt: normal<br>• Congenital Anomalies 25–50%<br>○ Short Stature<br>○ Skeletal (Tri phalangeal thumb)<br>○ Craniofacial anomalies<br>○ Renal Anomalies<br>• CBC:– Low Hb with high MCV.<br>• Genetic study, BMA.<br>Rx<br>○ PRBC Transfusion<br>○ Corticosteroid<br>○ BMT or SCT .<br>• Increase Risk of Leukemia |

**Transient Erythroblastopenia of childhood (TEC):– (imp DD to DBA)**

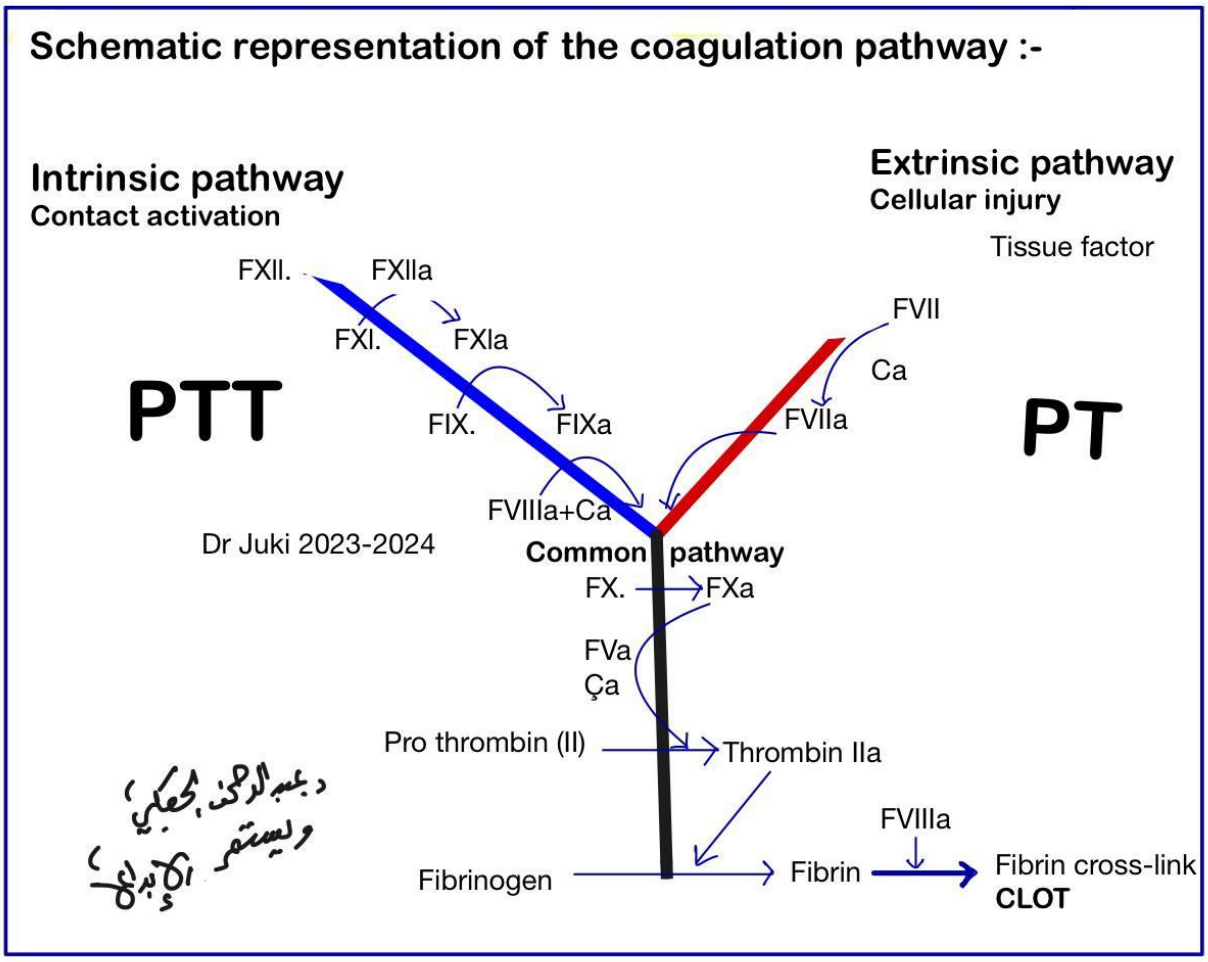
1. Age [6m – 5 years], usually triggered by viral infection.
2. Low Hb (only red cell defect)– No anomalies – normal MCV
3. Spontaneous recovery usually within several weeks.
4. Follow up – Transfusion for symptomatic anemia.

# Bleeding disorders

## Hemostasis:

- Is the mechanism of stoppage bleeding after injury of blood VS
- 3 components:
  1. Blood VS —>V C
  2. Platelet Adhesion
  3. Clotting factors

## "Coagulation Cascade"



## Bleeding disorders either:

1. Platelets disorder (inherited or acquired )
2. Coagulation disorders (inherited or acquired)

## Difference between Plt disorder & coagulation disorders:مهم

| | Plt disorders | Coagulation disorders |
|---|---|---|
| History | mainly mucous membrane bleeding or skin rash<br>1. Epistaxis<br>2. Gum bleeds | Deep tissue bleeding into muscles and joints "Hemophilia " |
| Examination | Petechiae – purpura–bruises | Ms Hematoma – Hemarthrosis–Bruises |
| Relation to trauma | • immediate<br>• Usually spontaneous | • usually delayed " oozing "<br>• Usually traumatic |
| Laboratory | 1. Plt count low<br>2. Abnormal function<br>3. Bleeding time | 1. PT.<br>2. PTT .<br>3. Thrombin time or specific factors level . |



## Common tests in evaluating bleeding disorders

1. **CBC** :–Platelet count
2. **Prothrombin time PT**:– it test extrinsic pathway "measures the activity of factors II, V, VII and X " (normal: 12–15 sec)
3. **Partial thromboplastin time APTT**:– it test intrinsic pathway " – measures the activity of factors II, V, VIII, IX, X, XIand XII " (normal :25–40% sec)
4. **Thrombin time** :– test for deficiency or dysfunction of fibrinogen.
5. D **dimers** :to test for fibrin degradation products
6. **Bleeding time** :– used to investigate platelet disorders, (nowadays on longer used & unreliable, It has been replaced by in vitro tests of **platelet function on a platelet function analyzer**).
7. Screening :– including renal and **liver function tests**.

Note: In the **neonate**, the levels of all clotting factors except factor VIII (FVIII) and fibrinogen are lower/ preterm infants have even lower levels

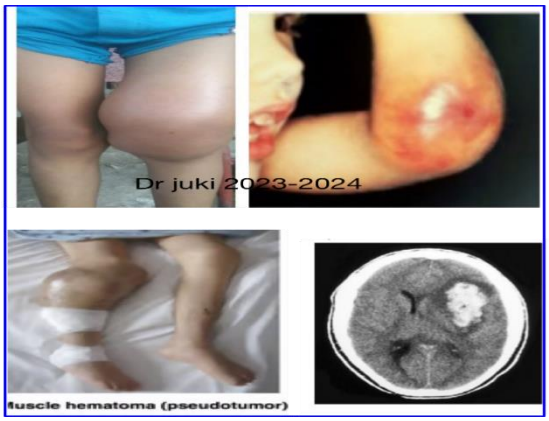
# Coagulation Disorders

## Classification

a. Inherited: **Hemophilia (A,B.C), vWD** & other coagulation factor deficiencies

b. Acquired: **vitamin K deficiency**(شرح في شابتر نيونيت), advanced liver disease, DIC, anticoagulants & acquired inhibitors of coagulation ( following Rx with F VIII or IX )

<table>
<tr><th></th><th>Hemophilia</th><th>Von Willebrand's Disease "VWD"</th></tr>
<tr><td>Mode of inheritance</td><td>XLR : ♂ only affected<br>: ♀ carrier<br>2/3 +ve F/H – 1/3 sporadic.</td><td>AD: ♂ = ♀<br>It is the most common hereditary bleeding disorder (1% of population)</td></tr>
<tr><td>Types</td><td>
<table>
<tr><th>Hemophilia A</th><th>Hemophilia B (Christmas disease)</th></tr>
<tr><td>1:5000 male births.,common<br>85%<br>Factor VIII ↓↓<br>C/P :sever</td><td>1:30,000 male births.<br>10–15%<br>factor IX ↓↓<br>C/P :milder</td></tr>
</table>
Hemophilia C : Factor XI ↓↓, very rare 2–3% ,Autosomal Recessive, most milder type or No bleeding.</td><td>1. Type I :– partial deficiency in VWF (commonest ,60–80%),mild & diagnosis late puberty.<br>2. Type II :–defect in VWF (qualitative)<br>3. Type III:–complete absence in VWF<br>• vWD (either a quantitative or qualitative deficiency of vWF).</td></tr>
<tr><td>Severity</td><td>is graded as severe, moderate, or mild, depending on the FVIII (or IX in hemophilia B) level
<table>
<tr><th>Factor VIII</th><th>Severity</th><th>Bleeding tendency</th></tr>
<tr><td><1%</td><td>Sever</td><td>Spontaneous joint /ms bleeds</td></tr>
<tr><td>1–5%</td><td>Moderate</td><td>Bleed after minor trauma</td></tr>
<tr><td>>5–40%</td><td>Mild</td><td>Bleed after surgery</td></tr>
</table></td><td>Function of VWF<br>1. Facilitates plt adhesion to damaged endothelium<br>2. Carrier of factor VIII —>Protecting if from inactivation and clearance</td></tr>
</table>

| | | |
|---|---|---|
| **Clinical picture** | 1. **Neonatal period** (40%).. <br>• Bleeding per umbilical stump <br>• Bleeding post circumcision <br>• prolonged oozing from heel stick and venipuncture sites. <br>• Intracranial Hemorrhage. <br>2. **Childhood** (start Crawl– Walk)? <br>• Recurrent bleeding into joints or muscle (**Hemarthrosis** :hallmark!) <br>• Other bleeding – Hematuria, epistaxis, bleeding gums, post–tooth extraction bleeding. <br>• Intracranial Hemorrhage. <br>   | • Mucosal bleeding as epistaxis,menorrhagia <br>• Excessive prolonged bleeding after surgery as circumcision <br>• Bruising <br>• Large hematoma – hemarthrosis "very rare" |
| **Diagnosis** | 1. CBC :normal <br>2. PT :Normal <br>3. PTT :Prolonged <br>4. Bleeding time :normal <br>5. Specific factor (low) : VIII level or IX level according to type of hemophilia <br>6. Prenatal diagnosis: DNA study | 1. CBC :–Normal <br>2. PT: Normal <br>3. PTT :Prolonged <br>4. Bleeding time :Prolonged <br>5. vWF antigen & vWF activity (Ristocetin cofactor activity RiCoF) :Low |
| **Treatment** | **Management of bleeding :– Hemophilia A** <br>• Recombinant factor VIII: supplied as powder (Recombinant DNA technology), 8–12–hourly <br>• Dose of factor VIII :0.5 X BW X [% desired in plasma F VIII] | • Desmopressin (DDAVP): Lt release of vWF from the endothelium —> + \|vWF & F VIII. <br>• Recombinant factor VIII :used in severe types of vWD |

| | | |
|---|---|---|
| | • % Desired :– 30% of normal is sufficient to treat minor bleeds and simple joint bleeds. Major surgery or life–threatening bleeds require the level to be raised to 100% and then maintained at 30% to 50% for up to 2 weeks to prevent 2ry Hge.<br>• FFP or Cryoprecipitate<br>• Desmopressin (DDAVP) is used in mild hemophilia A (+ endogenous FVIII &vWF),<br>• Antifibrinolytic therapy(Tranexamic acid ) in mucosal bleeding (epistaxis, mouth, dental…)<br>• TTT of hemarthrosis (rest,cold spong ,analgesia,FVIII then physiotherapy).<br>**Prophylaxis**<br>• Factor VIII prophylactic therapy : in **severe** cases to prevent spontaneous joint bleeding.<br>• It is given every 2–3 /wk (IV) to maintain factor VIII level > 2%.<br>• Primary prophylaxis usually begins at age 2 to 3 yrs.<br>**General measures**<br>oCareful observation & psychological support<br>o Avoid trauma, salicylates & NSAIDs<br>o Avoid IM injection.<br><br>• **Note** :management and prophylaxis in hemophilia B (recombinant FIX).<br>  <br> | • Cryoprecipitate is no longer used to treat vWD as it has not undergone viral inactivation.<br>• Avoid trauma, salicylates & NSAIDs& IM injection. |

| | | |
|---|---|---|
| | • **Note**:Desmopressin & Cryoprecipitate are ineffective in hemophilia B & C.<br><br>**Box 23.5** Complications of treatment of haemophilia<br>***Inhibitors, i.e. antibodies to FVIII or FIX***<br>• Develop in 5% to 20%<br>• Reduce or completely inhibit the effect of treatment<br>• Require the use of very high doses of factor VIII or bypassing agents (e.g. FVIIa) for treating bleeding<br>• May be amenable to immune tolerance induction<br>***Transfusion-transmitted infections***<br>• Hepatitis A, B, and C<br>• HIV<br>• Other, e.g. Prions, parvovirus B19<br>***Vascular access***<br>• Peripheral veins – may be difficult to cannulate<br>• Central venous access devices may become infected or thrombosed | |

# "Thrombocytopenia"

**Thrombocytopenia: is Plt count <150x10⁹/L. (N 150,000 – 45000)**

- **Mild** thrombocytopenia: Plt count 50–150x10⁹/L.(low risk of bleeding unless there is a major operation or severe trauma).
- **Mod** thrombocytopenia: plt count 20 – 50x10⁹/L.(risk of excess bleeding during operations or trauma ).
- **Severe** thrombocytopenia: Plt count <20x10⁹/L (risk of spontaneous bleeding.)

## Causes of thrombocytopenia (MCQ)

| 1. Increase Plt Destruction | 2. Decrease Plt Production |
|---|---|
| **Immune**<br>1. ITP "Idiopathic thrombocytopenic purpura"<br>2. SLE<br>3. Allo immune neonatal thrombocytopenia<br>4. Evan syndrome.<br>**Non Immune**<br>1. HUS<br>2. TTP (Thrombotic thrombocytopenic purpura).<br>3. DIC.<br>4. MAHA "CHD"<br>5. Giant hemangioma "Merritt – kasabach syndrome"<br>6. Hypersplenism | **Congenital**<br>1. Fanconi anemia<br>2. Wiskott – Aldrich syndrome<br>3. TAR syndrome "Thrombocytopenia – Absent radius "<br>4. Bernard – Souldier syndrome<br>**Acquired**<br>1. Aplastic anemia<br>2. BM infiltration (Leukemia)<br>3. Drug induced<br>4. Sepsis |

## Purpura :–

**Definition:** it is a small hemorrhage into the superficial layers of the skin producing areas of purple discoloration which do not blanch on pressure. Minute spots of 1–2 mm are called petechiae and larger areas of 1–2 cm are called ecchymoses

**Classification**

1. Thrombocytopenic purpura (Platelet count < 150.000/ mm') كما ذكر في الجدول فوق
2. Non–thrombocytopenic purpura (**Vascular or platelet dysfunction=Thrombasthenia**")

| Vascular causes (Normal Plt count) | Platelet dysfunction (Normal Plt count) |
|---|---|
| • **Congenital** :Marfan's syndrome – Ehlers – Danlos Syndrome<br>• **Acquired** : HSP – Meningococcal – scurvy . | **Congenital**: Rare disorders, e.g. Glanzmann syndrome .<br>**Acquired**: Uraemia (CKD), cardiopulmonary bypass. |

# "Idiopathic thrombocytopenic Purpura "ITP"

## "Immune thrombocytopenia"

- ITP Is the **commonest** cause of thrombocytopenia in childhood
- **Causes**: Idiopathic {Immune – mediated destruction of circulating Plt due to antiplatelet IgG autoantibodies}
- **Pathogenesis**:–Antibody–coated platelets are destructed in the splenic sinusoids (recognized by Fc receptors).
- **Age**:– Between 2–10 years (Peak 2–5 years)
- Usually follows a viral infection (H/O URTI 1–3 weeks) Or vaccination.

## Clinical picture:

1. Petechiae – purpura – superficial bruising
2. Bleeding
   - Epistaxis – gum bleeding
   - Profuse bleeding (uncommon)
   - Intracranial bleeding is serious but rare [0.1–0.5%]
3. Patients usually do well , No Organomegaly , No Lymphadenopathy .

**Diagnosis:** [Usually based on clinical presentation + Plt count] , diagnosis of exclusion.

1. CBC (Hb N – WBC N – PH low <150) .
2. Bleeding time :Prolonged.
3. PT and PTT : Normal .
4. Blood smear : normal morphology of plt + decreased Plt number .
5. BM aspiration:Done only if you are in doubt to exclude leukemia – (BM picture in ITP : **increased megakaryocytes**)

Note :– Any **atypical** clinical features, such as the presence of **anemia, neutropenia, hepato–splenomegaly, or bone pain ,marked lymphadenopathy,blast cell in blood film** should do a bone marrow examination to exclude **acute leukemia or aplastic anemia**

6. Anti Plt Ab : No role in Diagnosis (Nonspecific)

**Treatment:** **≃80% of children the disease is acute, benign and self – limiting, usually remitting spontaneously within 6–8 weeks.**

1. Most cases can be managed at home
2. **Observation** – ovoid trauma & IM injection .
3. If severe bleeding or persistence bleeding or Plt <$10x10^9$/L:–
    1. Immunoglobulin IV 2g/Kg/single dose (DOC) : it block Fc receptors ↓ platelet destruction
    2. Prednisolone oral 1–2 mg/Kg/day for 2 weeks
    3. Anti D or Rituximab, immunosuppressive.

- **Platelet transfusion** & Splenectomy are contraindicated and should be avoided in Rx of **acute ITP** except in severe life threatening bleeding .

Definition of ITP based on duration:

- **Newly–diagnosed or acute ITP: Duration ≤ 3 months**
- Persistent ITP: Duration 3–6 months
- **Chronic ITP: Duration ≥ 6 months.**

- **Chronic ITP**
    - ITP persists ≥6 months.
    - ≃ 10 – 20 % of acute ITP .
    - Rx :– Observation, follow up in most cases .
        - thrombopoietin receptor agonists (TPO– RA) 1st line as eltrombopag or romiplostim , Rituximab (2nd line).
        - Splenectomy if failed medications.

- **Evan syndrome** :– Hemolytic Anemia + Immune thrombocytopenia .
- SLE screening should be performed in case chronic ITP .(imp cause of low plt)

# "Disseminated intravascular coagulopathy DIC"

- Is a disorder in which widespread activation of the coagulation mechanism is usually associated with shock "Bleeding + clotting manifestation may present"

## Causes:

| | | | | |
|---|---|---|---|---|
| 1-Sepsis ***<br>Meningococcemia<br>Rickettsia | 2-Tissue injury<br>Shock<br>Crush injury<br>Burns | 3-Malignancy | 4-Venom ,Toxin<br>Snake bites | 5-Neonatal<br>N Sepsis<br>N E C<br>Server RDS |

## C/P

- Clinical picture (Critically ill)
- Picture of the cause (sepsis, shock, GE. . .)
- Bleeding: Purpura, ecchymoses, puncture sites, internal hemorrhage
- Thrombosis: necrotic skin patches

## Diagnosis:

1. CBC : plt low ,Hb low .
2. PT- PTT-bleeding time : prolonged
3. Fibrinogen : low
4. D Dimer & FDP : High

## Treatment:

Is Challenging

1. ICU + Rx of shock + FFP + plt transfusion ,treatment of the cause.
2. Antithrombin and protein C
3. Heparin: Use is controversial .

# "Leukemia"

- Is the **commonest** tumor in pediatric age groups [account ≃25–30% of all childhood cancer]

## Epidemiology:

- Peak "**age 2–5 years**"
- Sex "**Boys**> girls"

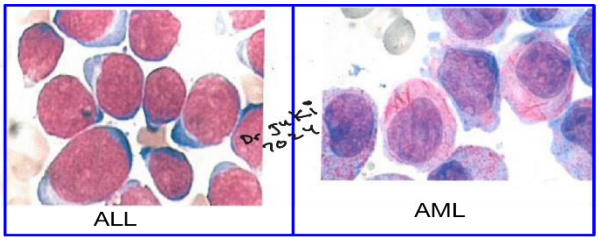
## Etiology:

The etiology of childhood leukemia is unknown and is probably multifactorial [Genetic – environmental play imp rules]

## Risk factors:

1. Genetic predisposition – +ve F/H (siblings of children with leukemia)
2. Environmental [Ionizing radiation – chemical Exposure]
3. Chromosomal [Down syndrome – Bloom syndrome]
4. Others [Fanconi anemia – NF1 –Wiskott – Aldrich syndrome – Ataxia telangiectasia – Diamond syndrome]

## Types:

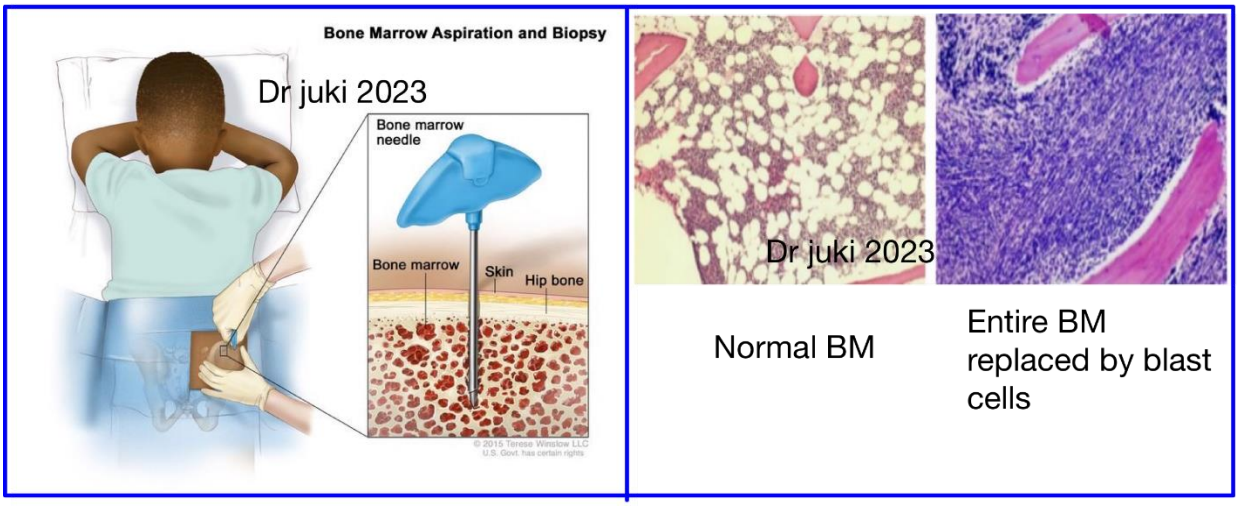
| Acute Leukemia 97% | Chronic Leukemia ≃ 3% |
|---|---|
| • **Acute lymphoblastic leukemia "ALL" 75–80% ***** <br>• Acute myeloid "Leukemia AML" 20–25% <br>   | • Chronic lymphocytic Leukemia "CLL" <br>• Chronic myelogenous Leukemia "CML" ( Ph +ve or Ph –ve ) <br>• Juvenile myelomonocytic Leukemia "Poor" |

## Clinical Picture:

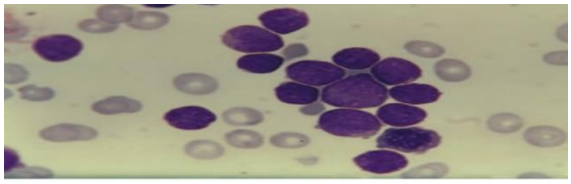
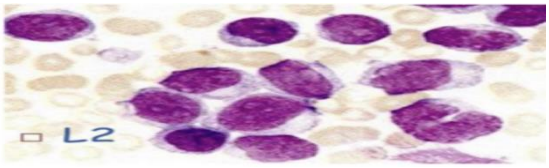
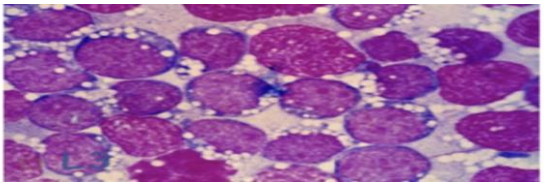
| General | BM Infiltration | RE infiltration | Other organ infiltration |
|---|---|---|---|
| Anorexia<br>Weight loss<br>Malaise<br>Irritability | **Anemia** (Pallor – Lethargy)<br>**Decrease** WBC [Infection – Fever]<br>**Decrease** Plt [Bruising – petechiae , bleeding]<br>**Bone Pain** – may be only & initial presentation.<br>(Note in AML :gum hyperplasia,chloroma,huge spleen) | Hepatosplenomegaly ≃70%<br>Lymphadenopathy ≃ 5 – 50% (mediastinal and SVC obstruction) | **CNS**: high ICP<br>Headache<br>Vomiting<br>Nerve palsies<br><br>**Testes**:painless swelling<br>**Priapism** |

## Investigation:

1. CBC :– Hb low , Plt low , WBC low– N – high .
2. **Blood Film** : Blast Cells
3. **Bone Marrow aspiration "Diagnostic" Blast Cells (>20–30%) ≃ 80–100%**




4. Other Test
   - Urea/ Electrolyte – Uric acid – LDH – LFT – Coagulation profile.
   - Chest XRay – U/S Abdomen – Echo – ECG
   - CSF examination
5. Special Bone marrow study for classification "ALL"

| Morphology "FAB Classification" | Immunophenotyping by flow cytometry |
|---|---|
| **L1: (Small cell–scanty Cytoplasm )Best prognosis**<br>  <br>**L2: (Larger cells –Larger cytoplasm – irregular )Poor prognosis**<br>  <br><br>**L3 :(Large Cells –vacuolated cytoplasm) Worst prognosis**<br>  | 1–**B cells** :–75%<br>• Early pre B cells **common ALL** (better prognosis)<br>• Mature<br>2–**T cells** :–15% Bad prognosis .<br><br>Also **Chromosomal** Test:<br>1–Number :– Hyperdiploidy (good ) but Hypodiploidy(poor).<br>2– Translocation Philadelphia (9:22 and 4:11) :Poor prognosis . |

## Treatment:

### 1. Supportive Rx

- Control infection
- Control bleeding
- Control Anemia
- Good Hydration – Allopurinol? To ovoid tumor lysis syndrome

### 2. Specific Rx "Chemotherapy"

**1–Induction chemotherapy (Induction of remission):** Prednisolone –Vincristine –L Asparaginase – adriamycin given for 4 weeks (>85% remision )
**CNS prophylaxis:** To prevent CNS Relapse– intrathecal methotrexate or Cyclophosphamide

2– **Consolidation phase:** to maintain remission with prevention of drug resistance, either same drugs of induction Or different drugs combination (better),for 4wks.

**3–Maintenance of remission:**Long term remission with prevention of relapse Duration: 2–3 years

**3–Bone marrow transplantation:**– if frequent relapse or poor response.

## Complication of ALL:–

1. **Tumor Lysis syndrome** :– Is a group of metabolic complications that occur before or after treatment of leukemia by chemotherapy.
   - Tumor cells very sensitive to chemotherapy
   - Rapid lysis :**hyper k– high uric acid – hyperphosphatemia– hypo Ca** .
   - Acute kidney injury (ARF ).
2. **Relapse** : common sites (MB – Tests – CNS)
   Rx :High – dose chemotherapy usually with total body irradiation and BMT
3. **Febrile neutropenia** (fever + neutropenia): Rx Broad spectrum Antibiotic IV
4. **Complication of Drugs**
   1. Steroids :(Cushing syndrome)
   2. Chemotherapy (adriamycin :cardiomyopathy) (methotrexate:stomatitis ,renal toxicity)(cyclophosphamide:Hgic cystitis +infertility ), liver toxicity ,alopecia.

5 .**Sepsis , Bleeding, DIC–Encephalitis**

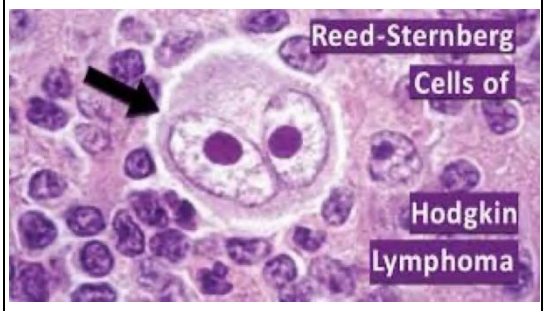
| Poor Prognostic Factors "ALL" High Risk ( MCQ ) |
|---|
| 1. Age: <1 year – ≥ 10 years<br>2. WBC (initial count) >50000/$mm^3$<br>3. Race :Black<br>4. CNS leukemia at diagnosis<br>5. Morphology FAB :L2 L3 – T Cell type<br>6. Sex : ♂<br>7. Cytogenetics: t(4:11) and t(9:22) Philadelphia chromosome and hypo diploidy<br>8. Response to induction therapy : slow or detectable Minimal residual disease after induction therapy. |

- **Congenital Leukemia** (leukemia diagnosed in 1st 6 weeks of life , rare – may be associated with Down syndrome).

- **% of Pediatric Tumor**
  1. Leukemia 28% (ALL***)
  2. CNS tumor 21%
  3. Lymphoma 11%
  4. Neuroblastoma 7.5%
  5. Wilms tumor 6%

# Lymphoma

- Lymphomas are **malignancies of the cells of the immune system**
- Lymphoma is the 3rd most common malignancy in children
- Lymphoma classified into
    3. Hodgkin disease (more in adolescence)
    4. Non Hodgkin Lymphoma (more in children)

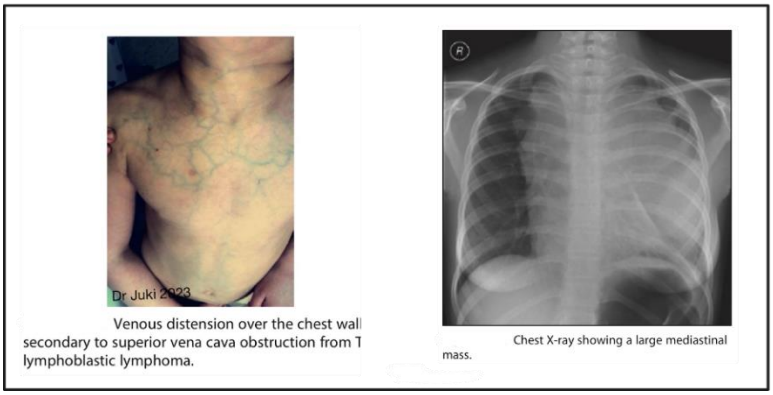
| | Hodgkin Lymphoma | Non – Hodgkin Lymphoma |
|---|---|---|
| Site | Essentially is nodal disease extra nodal sites (rare ) | Essentially is extra nodal disease (More common ) |
| Age | Bimodal – Adolescence and >50years | Childhood (% increase with age) |
| Sex | Increase ♂ (2:1 ♂ :♀) | Increase ♂ and white . |
| Pathologic Hallmark | Reed – Sternberg cells   | Not present |
| Clinical picture | 1. Lymphadenopathy painless – Film **cervical** –axillary – inguinal mediastinal (cough ,airways obstruction)<br>2. B Symptoms :<br>- Pel Ebstein Fever (>38°c)<br>- Night sweating<br>- Weight loss ,pruritus<br>3. Other [Anorexia – bone pain – Hepatosplenomegaly] | 1. Systemic symptoms fever – sweating – weight loss (rare)<br>2. Abdominal mass: intestinal obstruction,ascites , hepatosplenomegaly<br>3. Mediastinal mass: SVC obstruction presenting with dyspnoea, facial swelling and flushing venous distension in the neck and distended veins in the upper chest and arms.<br>4. Lymphadenopathy(B cell type)<br>5. BM infiltration : Failure<br>6. Join lesions [Burkitt Lymphoma] |
| Types | 1. Nodular sclerosing (40–50%)<br>2. Mixed cellularity (40%)<br>3. Lymphocyte predimonic<br>4. Lymphocyte depleted (L1 rare) | 1. Burkitt Lymphoma (B cell) EBV<br>2. Lymphocytic (T cell)<br>3. Large cell (B ,T cells) |

| | | |
|---|---|---|
| **Investigation** | 1. FBC ,ESR,LDH<br>2. Blood Film,BMA.<br>3. C XRay – U/S<br>4. Lymph node Biopsy<br>5. CT scan for staging . | 1. FBC ,ESR,LDH<br>2. Blood Film,BMA.<br>3. C XRay – U/S<br>4. Lymph node Biopsy<br>5. CT scan or MRI . |
| **Treatment** | **According to staging:–**<br>Chemotherapy ± Radiotherapy<br>• Positron emission tomography (PET) scanning is used to monitor treatment response. | According to staging<br>Chemotherapy + Radiotherapy<br>• survival rates of over 80% are expected for both T–cell and B–cell disease. |

**Burkitt Lymphoma** is a type of B–cell non–Hodgkin lymphoma and has three variants.

1. **Endemic (common)** : EBV (malaria endemic regions =africa) characteristically involves the jaw or other facial bone
2. Immunodeficiency– associated **Burkitt** lymphoma is usually associated with **HIV infection** or occurs in patients on immunosuppression post transplant.
3. **Sporadic** : in **Western** Word .

**Treatment is** with multi–agent chemotherapy.



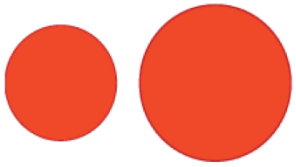
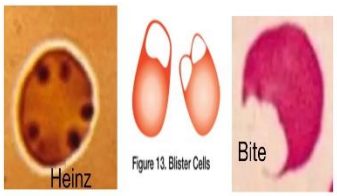
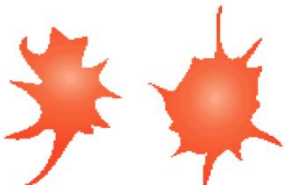
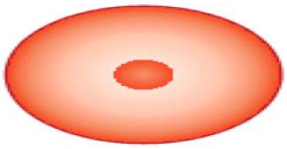
Venous distension over the chest wall secondary to superior vena cava obstruction from T lymphoblastic lymphoma.

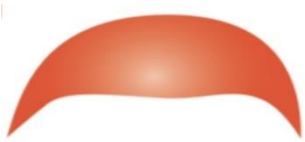
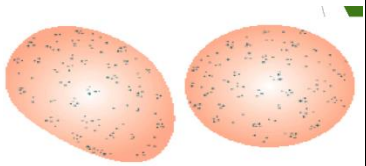
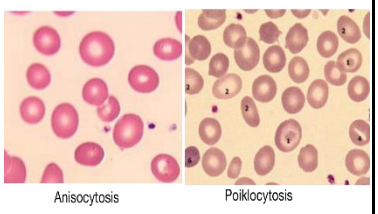
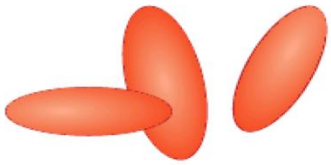

Chest X-ray showing a large mediastinal mass.



Burkitt lymphoma involving facial bones

# Blood film

| SHAPE | DD | BLOOD FILM |
|---|---|---|
| Spherocyte | • Hereditary Spherocytosis<br>• Autoimmune hemolytic anemias<br>• Thermal injury & BurnS. |  |
| Sickle cell | • Sickle cell anemia<br>• Sickle cell trait |  |
| Heinz bodies<br>Blister cell<br>Bite cell | • G6PD def (during attack only)<br>• Unstable hemoglobin (old & new)<br>• Oxidative drugs |  |
| Acanthocyte<br>(Spur RBC) | • Abetalipoproteinemia<br>• Vitamin E deficiency<br>• Liver disease |  |
| Target cell | • Hemoglobinopathies<br>• IDA<br>• Hyposplenism<br>• Hepatic diseases |  |

| | | |
|---|---|---|
| **Fragmented Red Cells (Schistocytes, Helmet Cells)** | MAHA,HUS<br>DIC |  |
| **Basophilic stippling** | • Chronic lead poisoning<br>• IDA<br>• Chronic hemolytic anemias |  |
| **Howell Jolly bodies** | • Megaloblastic anemia<br>• Hyposplenism,<br>• post splenectomy<br>• Dyserythropoietic anemia |  |
| **Anisocytosis(size)<br>Poikilocytosis(shape)** | • in Iron def &<br>• Hemolytic anemias |  <br>Anisocytosis Poiklocytosis |
| **Ovalocytes (Elliptocytes)** | • Hereditary elliptocytosis<br>• Renal and liver diseases<br>• Vitamin B12 deficiency<br>• Myelodysplasia |  |

## References


1– Nelson textbook of pediatrics 22nd edition.

2– Illustrated textbook of paediatrics 6th edition

3– Nelson essentials of pediatrics 9th edition.

4– www.uptodate.com

5_ www.medscape.com



تنبيه:

هذا العمل تم بعد جهد كبير من قبل الدكتور الاخصائي: عبدالرحمن الجعكي، ويعتبر ثمرة سنين من خبرة ممارسة طب الأطفال ، كل ذلك كان ممزوج بحنكة العطاء الاكاديمي للمادة

لهذا لايجوز نسخ أو اقتباس أي جزء من المحتوى الآ بعد التواصل مع الناشر: الدكتور عبد الرحمن الجعكي.